

ActInf Textbook Group ~ Cohort 6 ~ Session 4 (Chapter 2, Part 1) ~ 2/19/20

TextbookGroup/ParrPezzuloFriston2022/Cohort_6 | Cohort 6 ~ Session 4 (Chapter 2, Part 1) ~ 2/19/2024 | 57:15

all right welcome back cohort 6 it's 219
and we're in our first discussion of
chapter 2 so Oli thank you for
facilitating and kick it off and then
let's see what we
go okay thank you uh all right so in
chapter
two uh we delve into I mean uh the main
uh compon one of the main components of
active inference Theory uh namely uh how
to formulate action and perception uh
based on variational approaches so for
perception uh the optimization parameter
that needs to be minimized is the
variational free energy and for the uh
action counterpart the similar parameter
would be uh the expected free energy uh
so basically uh in this chapter we
encounter two of the central equations
of active inference Theory uh which are
equations 2.5 and
2.6 uh one for the variation of free
energy which
formulates how uh the agent or the
cognitive agent
optimizes uh the variation free energy
as its minimization parameter and
similarly in equation uh
2.6 uh we see the same uh same mechanism
as applied to a minimization of the
expected free energy in the case of
action uh but before uh going into the
details of uh what those equations show
and how how they work uh in the context
of active inference it might be helpful

to just uh go through the whole chapter

um I mean

uh in case uh there's

any there's I mean any Gap in the

understanding of what the whole picture

here is so uh

first uh as the title of the section 2.2

suggest perception as inference is one

of the motors of active inference

because uh as we know sometimes uh in

the computational or the traditional

cognitive science uh we treat perception

as uh simple information processing um

mechanism but here in active inference

perception uh is treated as a kind of

inference and not just symbol processing

or information processing approach and

the way to do that is uh to formulate it

in terms of uh basian theor uh base

Theory

so uh in box

2.1 uh there

is this fundamental rule of probability

about I mean sumon product rules which

is which are used uh in base Theory and

then in equation 2 .1 uh The Familiar uh

base theorem

uh uh is introduced and then uh through

an

example in figure

2.1 it shows how exactly priors and

beliefs are formulated and expressed in

base

theorem and later by

using uh by by using those summon

product rules uh we can write the

posteriors uh and specifically the

posteriors in the denominator as a kind

of marginalization over uh over the probabilities of observations so that's basically what's been done in equation 2.2 and then uh here on page 20 there's this a useful distinction between the notion of surprise as applied in Psychology or at least folk psychology and the notion of surprise an active inference or namely the basian surprise sometimes it's called surprisal surprisal function uh so uh basically the difference between those two concepts are uh they are obviously related but they're not actually identical so when we talk about psychological surprise it connotes something about the effective uh notion of that surprise I mean uh we need to be uh we need to be affected by uh By An Unexpected observation in order for us to be surprised but here when we talk about basium surprise it's much more formal and it only describes uh the degree or the extent of the unexpected event regardless of how effectively it influences us as human observers or cognitive observers so it's just basically um uh something very formal and statistical so um one other key movements of active inference formulation is the use of callback lier Divergence uh for the comparison uh uh I mean uh for the comparison between the prediction and The observed uh data so uh the the justification to use callback lier Divergence comes from uh neon Pearson Lemma uh which I don't think is stated explicitly in the text

because uh in uh non Pearson Lema uh it states that the most efficient statistic to compare two distributions is the logarithm of their likelihood ratios so basically that's what callback lier Divergence is it's basically the uh um it's just um uh the comparison between the logarithms of the of the likelihood ratio it's the ratio between two uh likelihoods so uh here on page 20 we see the definition of callback lier Divergence in equation 2.3 and then it's applied to uh the example from figure 2.1 in order to calculate uh the difference between uh the posteriors and the priors uh from that example uh and then okay so here we comes we come to an interesting discussion about the difference between biological I mean about biological inference and optimality so uh in this case uh optimality uh can be I mean in active inference literature optimality is defined in two distinct way ways so here in this book optimality is used um as a as a as a cuss function uh that needs to be optimized or minimized so but there's another U that's basically called basian optimality but there's another notion of optimality uh which is called Janes optimality uh which is used in uh basian mechanics in order to calculate the dual aspect of active inference namely cmap

or constraint maximum entropy principle
uh which is the direct mirror uh of or
uh the symmetrical Counterpoint of
fe uh but without going into much
details about how that works and how
that relates to basian inference here
when the word optimality is used in this
book uh it's almost
always uh it almost always means basian
optimality and not Janes
optimality uh okay so uh the view of
active
inference uh
here we treat biological organisms as a
kind of machines uh that do uh that
undertake these um basian optimality in
order to optimize uh the surprisal or
minimize the variation of free energy or
expected free energy and the way they do
that is through a
tripartite um model namely uh the
generative process and generative model
and markup blanket that kind of acts as
an interface between the generative
model and the generative process so very
briefly generative process is what
happens in the environment uh I mean
external to the agent and the generative
model is uh what the
organism
uh
and perceive about the environment or at
least uh anticipate the
environment the data or the latent uh
hidden states of the or latent states of
the environment and uh it constitutes uh
its generative model based on those
processes uh and the way it interacts

with uh the generative process is
through Mark of blanket uh which will
come to uh much more detailed in chapter
3 so here on page 23 we can see this
tripartite um model in figure

2.2 and uh

then as again as the symmetrical
counterpart for perception as action we
can

formulate that mechanism similarly for
the action namely action as inference
and uh the only thing that needs to be
changed in equation

2.5 is the substitution of um is a
substitution of um I mean variation free
energy with the expected free energy so
basically uh those two equations are uh
related to each other and they're
somehow actually the mirror image of
each other or counterparts to each other
so um yeah that's basically uh I mean
the rest of the the rest of the chapters
from 2.5 unto the end of the chapter is
devoted to the derivation of those
Central equations uh which we can talk
about a little bit more detail uh this
week or the next week if you

will thank you Oli awesome

summary does anyone want

to like what's a part that they like to
to or a question or a thought one that
was written in the questions that we can
go to or a new

one I have a question in terms of uh
self assessment so does it in terms of
like creating almost like an atlas in
terms of like say if you utilizing like
a a flow mechanic in terms of active

inference so if you're in a state where you know you're working on a problem for example um using a tool and you've arrived at a certain point how to then map back without consciously interrupting um your Flow

State um what's the best way to kind of map backwards so you can extrapolate any information forwards so people have questions you know on how you arrived at a certain conclusion how you might be able to uh clarify if that makes sense there's a lot of that question like

um know how and know that so is it about knowing where you've gotten to or knowing how to navigate forwards or backwards those might be all the same answer or it could be all different answers and going you know swimming up the river isn't going to help you swim back down the river or there's a path on the side of the river um you might be able to use just passive recording like recording the screen grab or something like that so that certain parts could be checkpointed and and just kind of locked in I considered like OBX in that sense um but I was just thinking in terms of storage um you know if there's a way around that too in terms of like can you do a direct correlational map kind of like a neural net to go backwards through that you could build out and then apply forwards as well well on the kind of um incremental Improvement you could uh use a time-lapse video in OB yes like you

don't need to take a HD 30 frames per second you could do like a time lapse and then to but to kind of bring it to the generative model side you could come to develop your own model of perception and action in the interface with that tool and then by building out your model of perception and action with that tool you might arrive at a very distilled very reduced parameter description

yeah I mean that's how I'm kind of considering it like how can I build nodes backwards that might give me inference forwards with you know less energy input in the future or in the present like can you build uh neural nodes that would you know reproduce the same kind of U predictive result by examining um you know those priors and how they changed or branched from the

origin well I'll try to give one more active thought on on that like so we're looking to be building some generative models something like figure 4.3 so a huge amount of the actual action selection dependent causal unfolding of the world is basically all loaded into B Because D is just a prior a is just a mapping between observations and hidden States so basically everything about like how one's actions in the world change it going forward is the expected free energy equation 2.6 B Matrix here so let's just say that you were um just doing a a mouse movement tool um one way to kind of engineer that

would be try to um you know de memorize
or try to reconstruct long sequences of
planned Mouse movements another one
might be just hyper
locally understanding like the gradients
around the mouse location or around the
attention very
locally and then trying to determine
that um attention landscape or that
affordance landscape and then try to um
either find yourself within or create
new can iions of like canalized
attention where to go from point A to
point B it it may be just as simple as
just sliding downhill rather than
needing to like
invoke um symbolic
cues
okay it makes sense I'm just trying to
think I guess it's a little bit more
granular but I was considering more um I
mean depending so personally for myself
I would like to build something I guess
that's more eye tracking because I do
that more I think than Mouse tracking
um but that's a different conversation
for a different day
so I guess it would still be you could
do XY parameterization in that same
sense
but sorry go and if I understood you
correctly uh there is another uh related
uh sub theory of active imprints uh
called branching time active imprint uh
which kind of an attempt to improve the
efficiency of active inference
computation uh by using basian filtering
uh and um so instead of the iteration of

the update equations uh they uh
alternate between integration of the
evidence and the prediction of future
States

so it significantly improves improves
the and U also reduces the complexity
class of the of the algorithms and uh it
improves the benchmarks uh significantly
so uh yeah branching time active
inference is a direct attempt to uh
somehow do those kinds of computations
with much less energy and with much more
efficiency as well okay great thanks

I'll review

that here's kind of a
core image with the two ways that the
discrepancy is addressed or resolved
with like change your mind or change the
world like it's too warm in the room you
know there's a plus it's plus three
degrees too

warm either that can come to be
explained Away by updating belief that
that's the temperature or that can be
changed materially or you know the
tension could be held there but these
are kind of the options when there is a
Divergence and then how agents and the
environment interact um

Thomas oh no I have a question so it's
um oh yeah go for it oh yeah just in
general I didn't want to interrupt I
wasn't meant to inter I'm not reason my
hand to intervene on what's being
proposed talk about right now um just's
a question my question was related to in
the policy section later part of chapter
2 the C parameter related to uh gpie so

the future expect 10 expectation part I was just curious like because that seems like a very important that I was curious I saw there was a that element is there and it seems like kind a surprising surprise uh that there's a uh preference such an important part to play in the I guess I assume the pragmatic value part of the equation uh yeah it is it's part of that section um so I I'd love to hear I expecting that probably talk more about in later chapters but that was just something I caught in my mind just more about that if there's some just clarification about where that's coming from and why that's there yeah Oli or anyone what is the preference variable how does it come into play and how does active inference's treatment of expected Ed and preferred observations have similarities and differences with for example reinforcement learning yeah uh so the preference variable I mean uh it's again one of the key moves um in active inference to encode preferences uh so uh for expected free energy I mean if we use the exact counterpart of equation 2.5 uh then uh I mean we would uh we wouldn't need uh to incorporate any other additional parameters uh I mean to encode preferences or anything like that but uh one of the uh I mean the dependence on policies uh can be dropped from the

pragmatic value term uh that's that's the idea behind uh the the encoding of the preferences because um although in uh some of the earlier papers of active inference uh it it's not incorporated into into the equations uh I mean but uh in recent papers uh they tend to clearly distinguish uh this form of the free energy namely the expected free energy from the variation free energy by incorporating that pragmatic uh value explicitly as conditioned on a preference variable so uh basically uh the way they do it is to try to again uh embed the whole equation um under um or let let me put it this way uh they U by encoding the preference the whole equation can be uh somehow uh uh written down as a kind of lower bound of uh another optimization parameter which is which has incorporated this C parameter or the preference parameter so uh basically what it means is that uh by replacing the true posterior with an approximate posterior uh this uh pragmatic value can be explicitly conditioned uh pragmatic value can be explicitly conditioned on um the the preference variable C so uh by flipping the the terms inside that expectation value uh it can be uh it can be treated as the exact opposite of uh the variational free energy without compromising uh I mean the symmetrical relationship between those two uh so but why the epistemic value term here is

important well because obviously
uh so take for instance if we're in a
dark room
uh there's this pous problem dark room
problem the mapping between the hidden
States and the and the observations
would be entirely ambiguous so uh the
best way to reduce this ambiguity is to
uh uh somehow uh seek out the
observations that uh reduce this
uncertainty by U maximizing the change
from prior to posterior belief after a
new observation or uh similarly uh by um
uh I mean reducing the uncertainty about
those hidden states by reducing the the
variation of free energy so uh by
applying the epistemic value here it uh
provid Ides a way to navigate that space
of possibilities when the agent tries to
uh maximize its certainty about the
posterior belief uh so yeah that's one
of the key differences between the
earlier active inference literature from
about up until uh
2021 and the later ones from 20
21 uh up to recent
papers thanks a little more on C so this
is the PDP partially observable Markov
decision process and every letter here
is like a statistical
distribution or a variable in the
program so defining the active inference
agent is defining the a b c and and d
and so on so this is like defining the
whole agent in its unfolding and so we
can look at each of the nodes as
statistical distributions or
parameterized and then also when there's

an edge which is like a causal relationship it's an influence so we can talk about that as a conditional value like conditioned upon this being the case what does this and so that's the vertical line here so how does active inference deal with goal oriented behavior so it's a big question but at this kind of minimal essential level the key move is to define a preference distribution explicitly over observations so in reward or reinforcement learning you'd have observations coming in like let's just say we're talking about body temperature and different body temperatures would be associated with how rewarding they are and then movement would be performed to move to more rewarding States and utility discounting and all these other kind of ad hoc um issues arising in contrast what happens in the kind of a preference orientation in active inference which doesn't need to exist at all you could remove have a flat preference distribution and just have purely epistemic Behavior or or you could have purely preference driven Behavior but what happens here is that the preferences which is hence the the origin of the joke preference slash expectation like what do you expect and prefer is going to happen how does it become a self-fulfilling prophecy because there's a coincidence of the preferred and the expected in self-fulfilling agent so then the preference is explicitly the

distribution of body temperatures so when I'm at 37 degrees it's not like I'm in the most rewarding State and then if I drop down to 36 I need to increase my reward back by going to 37 it's actually I'm more surprised by being at 36 than 37 and the ball rolls back downhill to the bottom of the bowl resting in the low surprise State the most preferred the most expected the bottom of the bowl rather than trying to Hill Climb a secondarily proposed reward function or utility function that's scaffolded on an observation distribution so observations are not secondarily evaluated there's just the definition of a preference over observations and that kind of grounds the model it's kind of like having it touch grass because otherwise how would the actions come to influence in this way that is control like um the the fundamental piece of the model and then just last piece here um here's the G equation 2.6 expected free energy and then this is just showing how when you remove or zero out certain components of this equation how you get other well-known forms like if you remove epistemic value all of 2 three four five so all you retain is this pragmatic value of future observations conditioned upon preferences you get expected utility Theory whereas if you drop pragmatic value and only have epistemic value you have novelty maximization Infomax Etc so

this is kind of like a
generalization on a bigger space of
decision-making possibilities that run
the gamut from arbitrarily pragmatic to
arbitrarily epistemic and all of their
mixtures

David yeah so I just had a question
related to um you said rolling back down
um the bow so in terms of I'm just
trying to think of it analogously um
wouldn't that be kind of return to
priors but that in that sense would you
bring your new experience back down into
the bowl therefore expanding the priors
as well or would it be combinational in
a certain
way or bringing that back out of that
lower state to that reward State and
then back down would that
transfer any difference between the two
states it's a great question there's
been some work on C learning preference
learning so the easy setting is if
preferences are a priori fixed then um
yes the pragmatic component of the agent
will just just like a moth for a flame
just like it will just keep trying to
get to see and so maybe for like
thermostat or something like a fixed C
makes sense um then you could have a
higher order control policy on C that's
one option or Andor you could have C
learning now C learning is a bit of a
challenge because if you just updated
your preferences to whatever was
happening you might as well not have
preferences like if you just kept on
reup dating what temperature you

preferred to just whatever temperature was in the room you might as well just be going with whatever is happening in the room okay so it's more more of a recomposition in that sense or

Reformation um can I add something uh in answer to your question sorry uh so okay uh if I mean uh the uh probability of observation conditioned on C uh is totally different from the prior over states because um prior over states only uh encode beliefs about the true states of the world uh irrespective of the preference or what is preferred but uh in this formulation uh we uh by using uh the condition I mean by using uh the preference value see the observations with higher probabilities are treated as more rewarding so it's a kind of reward function uh if you will right but uh again it's totally different from the prior over states uh and um also another point uh this kind of formulation enables u a kind of planning as inference which is distinct from both perception as inference and action as inference right because here uh the use of these probability distributions to encode preferences and policy values uh is a kind of um it brings the elements of action selection uh within the domain of Bayesian belief updating uh and the key paper uh on that is uh Kaplan and Friston's 2018 paper namely um let me yeah find the name of the paper it's

something about yeah it's planning and navigation as active inference so this seemingly simple trick of using uh the preference value here uh again enables another form of inference which is completely different from the previous ones as

well okay great thank

you

does anyone want to um ask a question or we can look at upvoted questions or we can look at any other question

a lot happens in two this is kind of the first technical chapter and it's the low road with the how

Peter hi there um so I did have one question about

the um the kind of like triart type model that's shown on page

23 um so I understand the idea that um I'm trying to sorry just use the the correct terms here

um so so there's there's two hidden states that are listed one that corresponds to the world and one that corresponds to uh like the agent's model I understand the the concept of why the hidden State associated with the process would be hidden because it's on the other side of like a marov blanket why would the hidden State associated with the model be hidden like what is that what's the sort of implication

there go for it Ellie

uh okay so the uh the implication of uh

using the hidden state in the model um I mean by by using u
x
wait I sto hearing
only uh okay can you hear me yeah yeah
go ahead
okay uh so by including X in the generative model uh it so the X asterisk is the true value of hidden State uh which is in the generative process right and in generative model the agent tries to model to or tries to perceive as accurately as possible the true value of hidden state in the generative process say but uh obviously it can't be done uh through direct perception or any kind of uh with any level of U complete certainty uh and it can only be done uh probabilistically so uh the reason on one side it's only X and on the other side it's X ASX is to uh distinguish between those two values of x one being the true one the other being the inferred one right uh so here the X as hidden state in the generator model is actually the inferred hidden state of the true value of hidden state in the generative process so I'm not sure if it's helpful or not but yeah I'll add a few more pieces um close synonyms first off there's many um kind of natural language assertions and descriptions that don't interfere with this essential particular partition um which one of which I'll explain but the hidden State also sometimes called an external state or a latent State um but it its external is

hidden from the internal via the blanket
and also there's a sense in which this
is hidden from the point of view of the
environment and that symmetry um here is
a kind of very fun um kind of Light
breaking post textbook Group which is
here's what we were just looking
at generative model generative process
but then
later we see um ramstead and others
using generative model as system level
description and then in this video and
transcript um Ali
basically asks that
specifically so suffice to say that when
you're working with
multi-agent simulations or where there's
agents communicating across an interface
it isn't always the most useful to be
talking about internal and external
because internal is always just
specifically from one reference point so
if you're talking about ecosystems of
shared intelligence
then the way that they're addressing
that with with no fundamental change in
the actual topology of the model which
is really what defines the variables
um at least they are using more
generative model to describe the
total
constructed model representation so not
just not focusing on partitioning
generative model from process but again
these are different natural language
assertions on top of the same exact
formalities but it just can be a little
bit unclear because sometimes people are

talking about the environment as
generative process but then it's like
but if the environment is another agent
then doesn't it have a generative model
and what are we really talking about
well again the variables in the math or
the variables in the computer program
when you build it it wouldn't matter if
you described it one way or the other
per se but this more recent is just a
simpler more

inclusive um way to describe it however
it's also not the distinction between
generative model and process that the
textbook in insists on which is it's
accurate it's not like it's going to
mislead it's just that you'll also see
people talking about generative model in
a little bit of a broader
way David then

Thomas yeah so uh just in terms of
concept so I'm a little behind on some
of the uh terminology so I'm just trying
to grasp it a little bit more so I was
just thinking in terms of the Markoff
blanket

um as an opaque State versus a
transparent St so we updating are we
able to is it kind of like a
transference through that field state to
another higher state or is it you know
more of a transparency of the field
being able to glean context from you
know what is behind that and in that
sense you know going back to the model
that you were just showing if we are
let's say capable of generating
something that includes the model the

inferred State and the process wouldn't
we then create another blanket
encompassing the
whole and then how would you approach
that in terms of you know are you not
reproducing at the same time a new
state these are great questions yeah go
ahead Ellie okay sorry okay so the whole
idea behind markup blanket and the
justification for its use uh lies behind
the concept of sparse coupling so in a
complex multi-dimensional a agents or
systems uh there's a kind of sparse
coupling between the agent and the
environment in the sense that uh the
agent doesn't have any direct access to
the environment and uh the causal
relationship between uh the disturbances
of the environment or any kind of data
of the environment and uh its causal
effect uh on the internal state of of
the agent is uh completely nonlinear or
as friston himself would like to call
it's caused vicariously it's not direct
but uh it's kind of indirect in causal
INF influence on the internal states of
the agent so in that sense yes we can
say the markup blanket is kind of opaque
it's not transparent in the sense that
there is a
direct uh I mean transparent lens or
direct access to um environment to
external uh States and data but on the
other hand uh there's this notion of
basing lens which has developed uh
recently uh through category
the L yeah it is opaque they've explored
that in the phenomenology setting like

metser opaque states of mind and they've really mapped that closely to the blankie concept it's also opaque but but shines through that's why it's the holographic blanket and then you're totally right you can design or imagine special situations like where you know once you're dealing with metacognitive agents or counteract like you can end up with some very creative and and twisted settings and then it just like well you can make a map where San Francisco has you know an underground railroad and all these other things it's like okay and then those the map the map play can be infinite okay it's interesting thank you thank Thomas yeah the build on what Peter was asking originally with the hidden States um yeah we give it more is this a kind part of the distinction of AC and policy discussed in this chapter in that sense of what we just what you just said where the hypothetical counterfactual operations uh they seem to operate in in likelihoods as opposed to kind conf your probabilities uh so it seems like at least in my notes I had about that that the hidden States like basically the I guess expected X in the formulation is a segment or a trajectory of a hidden State over time and that's against or I guess uh conditionalized to P_i the condition the conditional trajectories on the policies so just

like there's some clarification about
that yes okay so let's think about Tilda
is the real temperature in the room
Through Time Tilda is just sequence
through time we're prospectively talking
about action now not all things
prospectively plan action so vfe 2.5
might be sufficient if we were talking
about like a rock but we're talking
about something that's actually engaging
in that counterfactual so the real
temperature in the room X Tilda through
time and then the observations Through
Time on the thermometer um so one of the
decompositions that we talk about the
most with expected free energy and the
one that aligns it the most with those
other models shown earlier of like
decision-making is this epistemic and
pragmatic value um distinction but that
decomposition is exactly equivalent to
first off the physics based energy minus
entropy form that's what makes it a free
energy but then these middle two forms
and these are linked um here is um how
ambiguous both have the same first term
which is how ambiguous what's the
entropy the policy dependent entropy of
the mapping between thermometer readings
and the temperature in the room so both
first terms are basically saying like um
policies are better when they're less
ambiguous in their sensory mapping and
then this second term is basically um
very similar looking the only difference
is here there's a y Tilda versus an X
Tilda
um because whatever our our action

selection dependence uncertainty about thermometer readings is if we had a perfectly accurate thermometer our risk over thermometer readings would be the same as the risk over temperature in the room and then to the extent that we have even a tiny tiny bit of noise in the thermometer we're always more unsure about the temperature in the room than we are about the thermometer readings because these are the the approximate Target of estimation of like preferences and then there's always further uncertainty Beyond observables so H how exactly complex World unfoldings get wrapped into this that is the the challenge of making the relevant model and what's interesting is that even if you write the equation and you u in your program and it calculates it this way you're also Computing a quantity or scoring measure that has these equivalent interpretations but this is usually how it's written but then that's equivalent to this physics based one this is the control theory based one and then here's kind of like a Investment Portfolio type one just wanted to say uh thanks for that explanation I think my misunderstanding uh with regards to X was thinking that hidden there meant something like hidden from the agent's perspective or like metacognitively inaccessible um so this

is very clarifying thank

you yeah like which es especially when
dealing with a complex cognitive entity
that like could have access or not to
its own State it's kind of like this

simple case is just we're writing the
software for a thermometer and air
conditioner and it just has Total Access
to the variables we're not going to
worry about the phenomenology or like
whether they something that's shielded
from itself or not um and and then again
in this minimal case like the aboutness
of X is exactly the the this x star the
real temperature in the room but you
could have a situation where the
temperature in the room causes the
lights to change

and then the agent has a has a model
where the Latent State it's inferring is
bitcoin price and it thinks that that's
a causal relationship with the lights
changing in the

room and like it will do its best with
the knobs that it has access to this is
kind of what variational methods are
given the the suite or the scope of the
knobs that it can change it will do its
best with a KL Divergence minimization
to to make the best

approximation but that doesn't even mean
that the aboutness of its model is like
remotely the

same like one case that's often talked
about well this could be a continuous
variable outside and then the agent
could just be saying too hot just right
or too cold so like the type of variable

and the aboutness of the variable can be
very
different

David yeah so I'm still trying to grasp
this and think I'm close but I'm not
sure so I'm kind of looking at this in
terms of analogously like unknown
unknowns and how that would update our
own priors um where you know we may not
be conscious I don't want to use the
word consciously but aware um that our
priors are being updated and then we do
externalize them

later I guess at what point do we then
re internalize them as

active good good question so let's just
say that we had a kind of a cognitively
like structurally cognitively fixed
robot um and uh then it doesn't even
have the

possibility to like uh like let's just
say it's inferring the the location of
one you know it's hunting one prey but
actually there's two out there or
there's none or something it's like it
will keep on doing what it can without
one with trying trying to find the
position of one um prey item and then um
higher and higher levels of
flexibility could could start to address
a greater diversity of real settings
externally but then that unknown unknown
is a really interesting question because
surprise is kind of like a um an
agnostic or or pronostic even um alarm
signal

because the an increase in Surprise due
to the the

temperature um reading that might be like kind of a Known Unknown like I can deal with this kind of temperature increase we've done it before or it could be something structurally different happening in the world that basically no action is going to be able to address right and then that's where like the structure learning comes into play and it's like okay well now let's propose new latent causes like wow the one cause model of temperature in the room is is is surprising I'm going to propose a two cause model of temperature and then like that the abductive process of proposing new accounts but but even that can still be beset with unknown unknowns right I guess I was approaching it in terms of and this I might go beyond the scope a little bit but in terms of sensory perception so like you know we have our five main senses but that's what's known to us right in terms of conceptualizing it there might be you know trending towards an infinite number of sensory inputs that you know the brain might process differently which we might find as you know our modeling gets better and we deal more with cybernetics and you know evolution in that sense aren't we kind of fixing the model in sense and not accounting for those extra sensory unknowns that might be incorporated later and would it require then a complete reconstruction of the model as we are conceptualizing

it

um with the metaphor of

LEGOs this is like the minimal Lego

tetrahedra four pieces this is the

minimal um you know pick it up and and

juggle with it Four Legos that are put

together that makes a total cognitive

simulation now if you want to deal with

where there's like higher order like

volitional or

adaptive structural changes to the

model just by sketching it out you can

see that the state space is going to

expand like Mega exponentially very

fast um so to do adaptive search or even

persistence in like structurally

volatile spaces is a very large question

so it w it wouldn't um unseat the basian

mechanics or make something simpler than

this but certainly you could make things

way larger than this graph that have a

lot of those kinds of like reflexive

structural modification and structure

learning it's just that in practice

first off not all of those um capacities

have been brought from the code and the

research or sorry from the research and

the math into the code so sometimes that

would need to be like implemented um ad

hoc in in a software package but we hope

to build that into the open source

packages and so on and then even if the

capacity has been translated from the

math into the code like structure

learning has been then there's often a

really really large optimization or

parameter sweep step because you're just

increasing the the state space of the

model so vastly okay so this is kind of
your gen state right for your your seed
growth basically so everything's going
to grow out of this despite you know
where it may grow too in that
sense like through the expansion we can
still go back to that base
model yeah like the the this uh
structure learning is like proposing
another latent cause and then asking
whether the the more um the higher model
complexity has sufficiently improved
accuracy and then there's basian model
reduction which is you take your model
and you ask like which of these
parameters is like the worst and which
one can I reduce out and so with this
kind of like plus one and minus one
parameter moves structure learning and
basing model reduction that's kind of
like that's like the AGI ACM kind of
dream SL meme would be you'd start with
a single state space and then there
would be a process that autonomously or
or in a scaffolded way that the agent
would basically propose alternatives to
its own model including like arbitrary
or higher order levels of whatever
cognitive State and then guiding that
kind of neuromorphic evolution again
it's an even larger State space yet it
would have like perhaps some of these um
qualities like parsimony or um like
ability to reproducibly develop
appropriately and so
on okay great thank
you well we'll
return next time for more low

road um up Vote or add more questions
and then if like you want to make sure
that it gets addressed because we didn't
look at too many of these just tag or
something and then at this time next
week that will be discreet Time chapter
7 in the cohort 5 which is kind of like
the models we were discussing today with
the

pdps hence why bouncing back and forth
and everything with the chapters is all
good so thank you all till next time
bye thanks guys have a good one